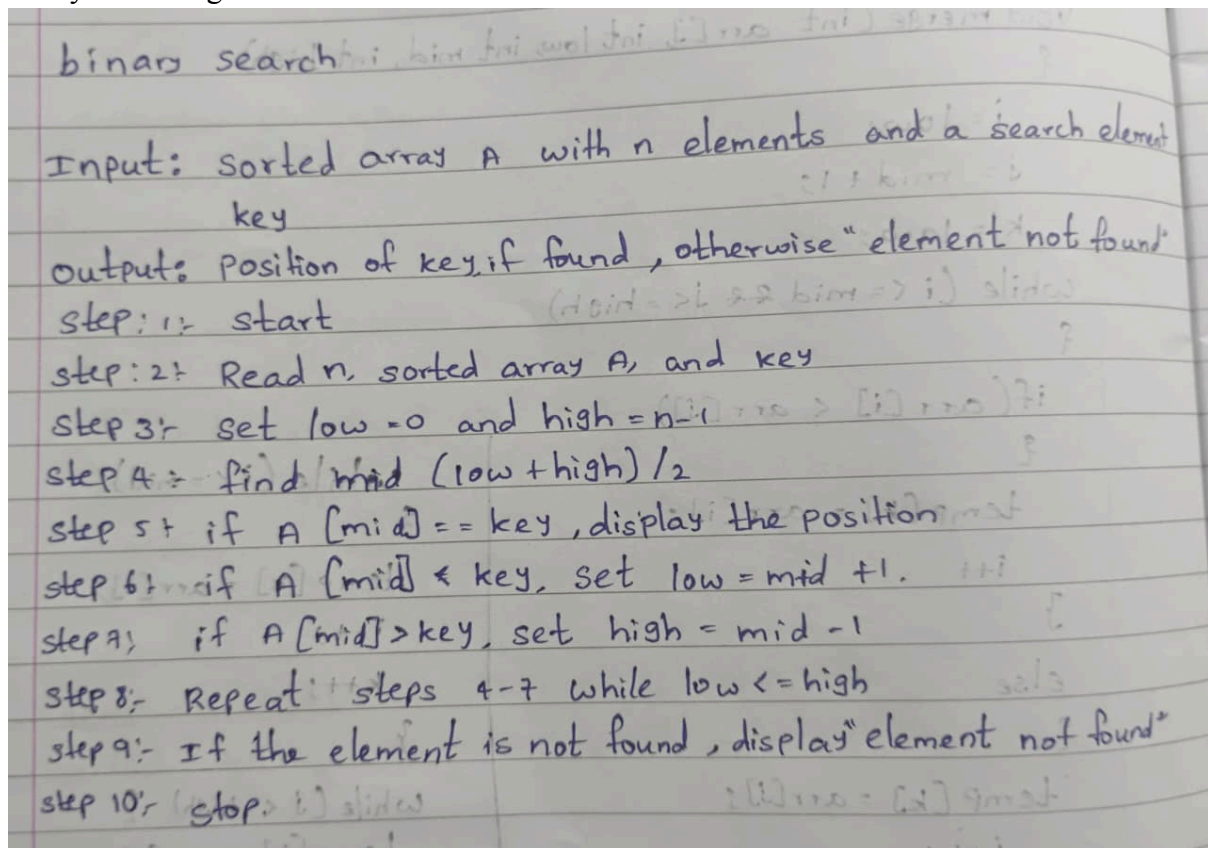


PRACTICAL - 2

AIM :Implementation and Time analysis of linear and binary search algorithm

binary search algorithm :



code :

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int n, arr[100], key;
    int low, high, mid;

    cout << "Enter size of array: ";
    cin >> n;
```



```
cout << "Enter sorted array elements: ";
for(int i = 0; i < n; i++)
    cin >> arr[i];

cout << "Enter element to search: ";
cin >> key;

low = 0;
high = n - 1;

while(low <= high)
{
    mid = (low + high) / 2;

    if(arr[mid] == key)
    {
        cout << "Element found at position " << mid + 1;
        return 0;
    }
    else if(arr[mid] < key)
    {
        low = mid + 1;
    }
    else
    {
        high = mid - 1;
    }
}

cout << "Element not found";

return 0;
}
```



Output

```
Enter size of array: 5
Enter sorted array elements: 2
5
7
35
89
Enter element to search: 5
Element found at position 2
```

time complexity : Step Count / Frequency Count Method)

```
Binary search

low = 0;      -1
high = n - 1; -1
while (low <= high) { O(logn)
    mid = (low + high) / 2; -1
    if (arr[mid] == key) -1
    { return mid;      -1
    } else if (arr[mid] < key) -1
    {
        low = mid + 1; } -1
    else {
        high = mid - 1; -1
    }
} return -1;      -1

T(n) = O(logn)
```



linear search algorithm :

linear search
Input: Array A with n elements and a search key
output: Position of key, otherwise "element not found"
step 1:- start
step 2:- Read n, array A, and key
step 3:- set i=0
step 4:- compare A[i] with key.
step 5:- if A[i] == key, display the position and go to step 8.
step 6:- increment i by 1 and repeat steps 4-5 until i < n
step 7:- Display "element not found."
step 8:- stop.

code :

```
#include <iostream>
using namespace std;
```

```
int main()
```

```
{
```

```
    int n, arr[100], key;
```

```
    cout << "Enter size of array: ";
```

```
    cin >> n;
```

```
    cout << "Enter array elements: ";
```

```
    for(int i = 0; i < n; i++)
```

```
        cin >> arr[i];
```

```
    cout << "Enter element to search: ";
```

```
    cin >> key;
```

```
    for(int i = 0; i < n; i++)
```

```
    {
```

```
    if(arr[i] == key)
    {
        cout << "Element found at position " << i + 1;
        return 0;
    }
}

cout << "Element not found";

return 0;
}
```

Output

```
Enter size of array: 5
Enter array elements: 34
67
23
65
1
Enter element to search: 24
Element not found
```

time complexity (Step Count / Frequency Count Method)



```
linear search

for (i=0, i<n; i++)    — n
{
    if (arr[i] == key) — 1
    {
        return i;    — 1
    }
}
return -1;    — 1

Tn = O(n)
```